Assignment 7

Salesforce: Overview and Website Deployment Guide

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1. Introduction to Salesforce

Salesforce is a cloud-based platform primarily used for Customer Relationship Management (CRM). It enables organizations to manage customer data, sales processes, marketing activities, and support services in a centralized environment.

Salesforce operates entirely on the cloud, eliminating the need for local installation and allowing access from any device with internet connectivity.

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2. Key Features of Salesforce

- Cloud-Based Platform: No installation required; accessible via browser
- Customer Data Management: Stores and manages customer information
- Application Development: Build custom applications using Salesforce tools
- Automation Tools: Workflow rules, Process Builder, and Flows
- Reporting and Analytics: Generate dashboards and reports
- Integration Support: Connect with third-party applications
- AppExchange Marketplace: Access to pre-built apps and extensions

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3. Core Concepts in Salesforce

Concept	Description
Organization (Org)	A Salesforce environment where data and applications reside
Object	Equivalent to a database table
Record	A single entry in an object (row)
Field	Attribute of a record (column)
Apex	Backend programming language similar to Java
Lightning	Modern user interface framework
Visualforce	Older UI framework used for building pages

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4. Salesforce Developer Edition

Salesforce provides a Developer Edition, which is a free environment for learning and development. It allows users to:

- Build and test applications
- Write and execute code
- Create custom objects and fields
- Deploy and manage applications

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5. Methods to Deploy a Website on Salesforce

Salesforce is not a traditional web hosting platform. Websites are built using Salesforce-specific technologies. The following are the main approaches:

5.1 Using Lightning Web Components (LWC)

This is the recommended modern approach for building user interfaces.

Steps:

1. Create a Developer Account
    - Sign up on the Salesforce Developer website and create a Developer Org
2. Install Required Tools
    - Visual Studio Code
    - Salesforce CLI
    - Salesforce Extension Pack
3. Create a Salesforce Project

```
sfdx force:project:create -n MyProject
cd MyProject
```

4. Authorize Salesforce Org

```
sfdx auth:web:login
```

5. Create a Lightning Web Component

```
sfdx force:lightning:component:create -n helloWorld -d force-app/main/default/lwc
```

6. Develop the Component
   Example:

```
<template>
  <h1>Hello World</h1>
</template>
```

What is this?

7. Deploy to Salesforce

```
sfdx force:source:push
```

8. Add Component to Page
    - Open Lightning App Builder
    - Drag and drop the component
    - Save and publish

5.2 Using Visualforce

Visualforce is an older method used to create web pages with server-side logic.

Steps:

1. Open Developer Console
2. Create a new Visualforce Page
3. Add the following code:

```
<apex:page>
  <h1>My First Website</h1>
</apex:page>
```

What is this?

4. Save and preview the page

5.3 Using Experience Cloud

Experience Cloud is used to build complete websites, portals, or communities.

Steps:

1. Navigate to Setup
2. Search for "Digital Experiences" and enable it
3. Go to "All Sites" and click "New"
4. Select a template
5. Customize using drag-and-drop tools
6. Publish the site

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6. Important Considerations

- Salesforce is not designed for hosting traditional web applications
- Technologies like Node.js, PHP, or standard MERN stack cannot be deployed directly
- Development must be done using Salesforce tools such as Apex and Lightning Components

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7. Use Cases for Salesforce Website Deployment

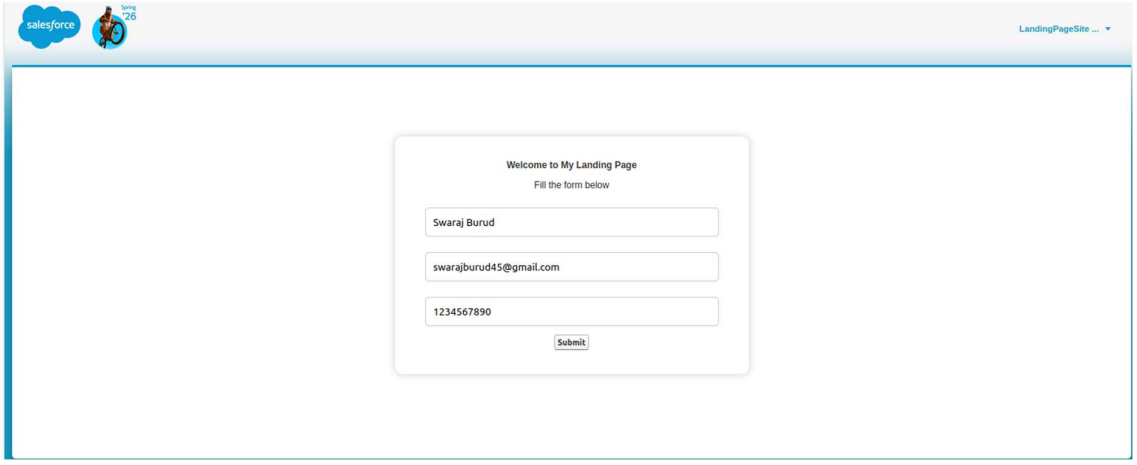
Suitable Scenarios:

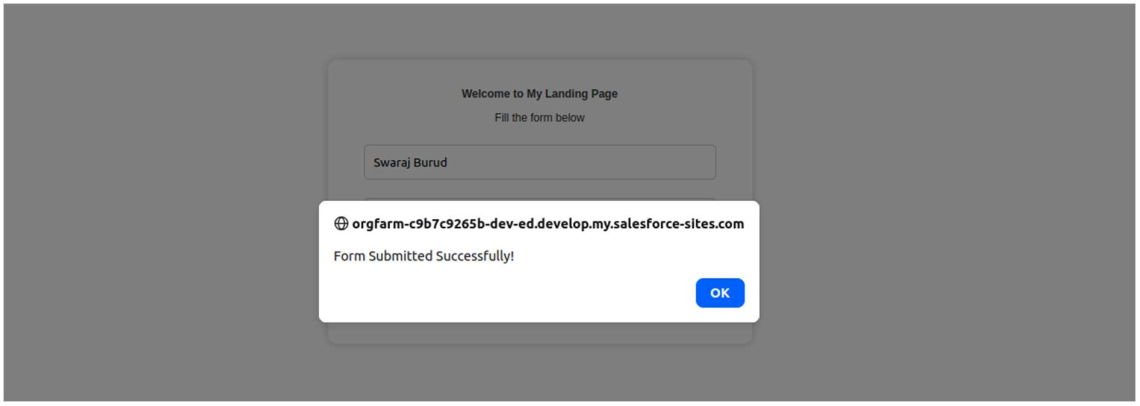
- Customer portals
- Business dashboards
- CRM-integrated applications
- Enterprise-level web applications

Not Suitable For:

- Static websites
- Personal portfolios
- Traditional full-stack applications

Output :





Conclusion

Salesforce is a powerful platform for building business-oriented applications rather than general-purpose websites. It provides robust tools for data management, automation, and user interface development. Website deployment in Salesforce is achieved through frameworks like Lightning Web Components, Visualforce, or Experience Cloud, depending on the project requirements.